Project Proposal

Linear vs. Nonlinear Dimensionality Reduction: An Empirical Study Across Face Recognition

Team: Outliers

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1. Problem Statement

A complete uncompressed audio recording involves an enormous amount of raw data—far more information than can practically be distinguished by the human ear. Audio compression is a method that works out which bits of the raw signal matter to the way a song sounds, and which parts can be thrown away. The file you get is a quarter of the size yet still sounds like the same song to a listener. Dimensionality reduction works similarly with data: take an image, and you have thousands of pixel values; take a genomic data set, and you have thousands of variables. All of this is far more raw information than the underlying pattern truly needs. Dimensionality reduction is the compression algorithm that removes excessive or non-essential detail but keeps the structure that counts.

To get a face recognized by a computer is a little like selecting between a pencil sketch and a sculpted clay bust. A sketch is a simple line drawing representing the basic contour of a face. Linear PCA represents faces as straight-line patterns in pixel data. It is speedy and does a good job on straight-on faces with even lighting. But actual features curve and tilt and change with pose and light, and a flat sketch cannot do full justice to a curling cheekbone or a reconfigured smile. Nonlinear approaches, like locally linear embedding or curvilinear component analysis, are more like a sculptor chiseling clay to conform to those true curves. The underlying dilemma is: do we want to use the more basic "sketch," the more complex "sculpture," or is there a way to tell beforehand which one a specific data set requires?

One might expect the sculpture, the more detailed nonlinear approach, to outperform the flat sketch. Yet testing real face data shows both perform almost equally well. This is significant because it means the added effort of nonlinear methods does not always pay off, challenging the assumption that more complex techniques are inherently better. Choosing the wrong method wastes computational resources without improving accuracy, and until now there has been no easy way to predict which method suits a given dataset.

To this end, we propose a *"curvature gauge"*—a measure of nonlinearity that shows how flat or curved a dataset really is, before a method is selected. Flat results support simple PCA; strongly curved results justify the increased cost of nonlinear approaches. This gives engineers, analysts, and researchers working on face recognition and image processing a way to analyze the structure of their data and then select the appropriate solution, saving time and computational power while keeping results precise.

2. Curvature Gauge

The curvature gauge is our main contribution. It is a number computed from a dataset, before any classifier or clustering method is run, that tells us how much a linear method (PCA) loses on that data compared with a nonlinear method (Isomap). We follow the residual-variance measure used by Tenenbaum, de Silva and Langford [5], whose Fig. 2 compares PCA, MDS and Isomap on this quantity. Residual variance shows what

a method is not able to capture of the data's distance structure at a given number of dimensions. Because PCA is linear and Isomap is nonlinear, comparing the two shows how much of the data's structure is nonlinear.

2.1 Step 1: Embed the data at several dimensions

Let X be the training images as pixel vectors. For every embedding dimension d in a grid from low to high ($d = 1, 2, \ldots, 10$, then $15, 20, \ldots, 70$, matching the 5 to 70 components used elsewhere in this proposal), we run both PCA and Isomap on X. Isomap uses a fixed neighborhood size K, chosen on the validation set. The gauge is computed on training data only. Points that are disconnected from the main neighborhood graph (infinite graph distance) are removed before computing, as in note 19 of the paper.

2.2 Step 2: Residual variance of each method

Following note 42 of the Isomap paper [5], the residual variance is one minus the squared linear correlation between a method's estimate of the data's pairwise distances and the pairwise Euclidean distances in its low-dimensional embedding, with the correlation taken over all pairs of points:

$$\mathrm{RV}_{\mathrm{PCA}}(d) = 1 - R^2\left(D_X, D^{\mathrm{PCA}}(d)\right), \tag{1}$$

$$\mathrm{RV}_{\mathrm{Iso}}(d) = 1 - R^2\left(D_G, D^{\mathrm{Iso}}(d)\right). \tag{2}$$

Here D_X is the matrix of Euclidean distances between the original images (the distance estimate used for PCA), D_G is the matrix of graph shortest-path (geodesic) distances from the Isomap neighborhood graph (the distance estimate used for Isomap), and $D_Y(d)$ is the matrix of Euclidean distances between points in the d-dimensional embedding recovered by each method. R is the standard linear correlation coefficient. A value near 0 means the embedding preserves the distance structure almost perfectly; a high value means structure is being lost.

2.3 Step 3: The gauge value

The gap between the two methods at dimension d is

$$\Delta(d) = \left|\mathrm{RV}_{\mathrm{PCA}}(d) - \mathrm{RV}_{\mathrm{Iso}}(d)\right|, \tag{3}$$

and the curvature gauge is the largest gap over all dimensions tested:

$$\mathrm{CG} = \max_{d} \Delta(d). \tag{4}$$

The sign of the difference is not important. Isomap has more freedom than PCA, so its residual variance will almost always be lower, and a gap will exist for every dataset. What matters is how large the gap is. We use the maximum over many d rather than a single d because the gap depends on the dimension: at very low d both methods may fail, and at high d PCA eventually catches up, so a single d can hide the curvature. In the Swiss roll (Fig. 2B of the paper), for example, PCA stays high until much larger d while Isomap drops to near zero at $d = 2$. In the face images (Fig. 2A), Isomap reaches near-zero residual variance by $d = 3$, while PCA needs about $d = 8$. Extra dimensions spent by PCA are the cost of curvature.

2.4 Step 4: Decision rule

We do not look at the gap alone; we also check the level of both residual variances. The rule is:

- **Large gap and a low Isomap residual variance** (CG $>$ τ_{gap}): the data are curved and the nonlinear method recovers structure that PCA cannot, so a nonlinear method is chosen.
- **Small gap** (CG $\leq$ τ_{gap}): linear and nonlinear methods lose about the same amount, so the data are effectively flat and simple PCA is chosen.
- **Both residual variances stay high even at large** d (above a threshold τ_{RV}): the data are intrinsically high-dimensional rather than curved, as for the handwritten "2"s (Fig. 2D of the paper). The gap is small here, but this does not mean the data are flat. We flag these cases separately and default to PCA as the cheaper method, since the nonlinear method gives no clear benefit.

The thresholds τ_{gap} and τ_{RV} are selected on the validation data only and are never tuned on the test set.

2.5 Step 5: How the gauge is tested

To test whether the gauge works, we compute it for many data conditions (each dataset, subsets of subjects or illumination conditions, and repeated random splits) and compare it with the accuracy improvement of the nonlinear methods over PCA measured on the same conditions. We report the Spearman correlation between CG and that improvement, and the method-selection accuracy of the decision rule against the best method found for each condition. This evaluation is what Objectives 2 and 3 depend on.

3. Objectives

- We will first reproduce and compare linear PCA with nonlinear dimensionality-reduction techniques, such as Kernel PCA, LLE, Isomap, CCA, and autoencoders, on the standard ORL and Yale face databases, using similar training, testing, and classification protocols as in previous work. We will extend this comparison to the Extended Yale B database, which contains photos of 38 subjects under a wide range of illumination conditions. We will also extend the analysis to an unsupervised setting by applying clustering methods to the reduced representations. This will allow us to examine when nonlinear methods provide an advantage over linear PCA in both classification and clustering.
- Existing methodologies do not provide a principled way to decide whether a dataset needs a nonlinear method or whether linear PCA will work just as well, which leaves researchers with costly trial-and-error testing. We will test the curvature gauge defined in the Curvature Gauge section. For each dataset, subset, and split, the gauge is the largest gap, over a range of dimensions d, between the residual variance of PCA and the residual variance of Isomap, computed with the definition in note 42 of Tenenbaum et al. (2000). We will check that the gauge value predicts how much the nonlinear methods improve over PCA in classification and clustering (measured by Spearman correlation), and we will use both residual variances together with the gap to separate curved data from data that are simply high-dimensional. This gives a data-driven, quantitative basis for choosing between linear and nonlinear techniques without running every method.

- Instead of treating linear PCA and nonlinear methods (Kernel PCA, LLE, Isomap, CCA, and autoencoders) as competing approaches, we will combine them into a single adaptive pipeline. The pipeline first computes the curvature gauge from Objective 2 on the training data. It then applies the decision rule: if the gap is large, the data are routed to a nonlinear method; if the gap is small, or if both residual variances stay high, the data are routed to PCA. This depends directly on the gauge, because the gauge value and its thresholds (selected on validation data) are the only input to the routing. We will evaluate the pipeline against always using PCA, always using the best nonlinear method, and an oracle that picks the best method for each setting, using method-selection accuracy and computational savings.

4. Dataset Description

In this work, three publicly available face-image datasets will be used, namely, the Olivetti Research Laboratory (ORL) database, the Yale Face Database and the Extended Yale B Face Database. The ORL dataset has 40 subjects and there are 10 photos for each participant, totaling 400 images. Each image has a resolution of 92×112 pixels and includes variations in lighting, facial expressions, and facial details. The Yale dataset contains 15 subjects with 11 images per subject, with images at a resolution of 243×320 pixels and variations in facial expression and lighting conditions.

The Extended Yale B Face Database will be included as a more challenging dataset because it contains images captured under a wider range of illumination conditions. The cropped frontal subset will be used for the primary experiments to maintain a consistent face-recognition setting across the datasets. If pose variation is examined as an additional experiment, the full version of the dataset may also be considered. The exact number of images used will depend on the selected subset and will be documented during the experimental stage.

Before applying dimensionality-reduction and machine-learning techniques, the datasets will be cleaned and preprocessed to ensure consistency across the experiments. Images will be converted to grayscale where necessary, cropped and resized to a common resolution, and transformed into numerical pixel vectors. Pixel intensities will be normalized to the range $[0, 1]$, and the feature vectors will be mean-centered using statistics calculated from the training data only. This prevents information from the validation or test data from influencing the training process.

Additional cleaning will be performed where necessary. In particular, some files in the Yale dataset use filename extensions that represent facial-expression information rather than a standard image extension. These files will be renamed and reorganized so that the subject identifier and facial-expression information are preserved while providing a consistent image format for processing. After preprocessing, the same preparation steps will be applied consistently across all datasets before dimensionality reduction, classification, and clustering.

5. Methodology

Our selected article compares linear and nonlinear dimensionality-reduction methods for supervised face recognition. We will reproduce this comparison and then focus on extending it to unsupervised learning. The reduced representations will be evaluated mainly through clustering, while supervised classification will be used for comparison.

5.1 Machine Learning Techniques

The analysis will begin with dimensionality reduction. PCA will be the main linear method and will be compared with Kernel PCA, LLE, Isomap, CCA, and a nonlinear autoencoder. The analysis will initially use 60 components, with additional experiments using about 5 to 70 components. The autoencoder will use an encoder-decoder structure with MSE reconstruction loss and early stopping. The reduced features will then be used for supervised classification with NN, soft k-NN, LDA, and SVM with linear and RBF kernels, and for unsupervised clustering with k-means and spectral clustering. Separately, we will compute the curvature gauge (see the Curvature Gauge section): PCA and Isomap are both run at a range of dimensions d on each dataset, residual variance is computed for each using the definition in note 42 of Tenenbaum et al. (2000), and the largest gap between the two curves, together with the level of both curves, is used to choose between PCA and a nonlinear method.

5.2 Evaluation Metrics

The reduced representations will first be evaluated based on how much information they preserve. For PCA, we will report explained variance and reconstruction error, while the autoencoder will be evaluated using RMSE. For the unsupervised analysis, clustering quality will be measured using the Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index. NMI and ARI will also be used to compare the clusters with the true subject labels.

For the supervised analysis, classification accuracy will be the main metric, reported as the mean and standard deviation across repeated data splits. Macro-F1 will be used as a secondary metric, and Δ accuracy will show the improvement of a nonlinear method over PCA. Computational cost will be measured using training time, projection time, and peak memory usage. Spearman correlation will be used to examine the curvature gauge, while the adaptive pipeline will be evaluated based on method-selection accuracy and computational savings.

The curvature gauge itself is evaluated with residual variance (note 42 of the Isomap paper) for PCA and Isomap at each dimension d, the maximum gap between them, and the Spearman correlation between the gauge value and the accuracy improvement of nonlinear methods over PCA. The thresholds are chosen on validation data.

5.3 Baseline Models

The main baseline will be PCA. For supervised learning, each nonlinear dimensionality-reduction method will be evaluated using the same classifier, data split, and reduced dimension as PCA, while for unsupervised learning, the same clustering methods will be applied to both PCA and nonlinear representations. A second baseline will use the original pixel features with a Nearest Neighbor classifier and no dimensionality reduction, and results on ORL and Yale will also be compared with those reported by Huang and Yin (2009). For the adaptive pipeline, we will compare its results with always using PCA, always using the best nonlinear method, and an oracle strategy representing the best method observed for each setting.

5.4 Data Splitting Strategy

The data will be divided into training, validation, and test sets. For ORL, 3 to 6 images per subject will be used for training, with the remaining images used for testing across 10 random splits. For Yale, 10 images per subject will be used for training and one condition will be held out for testing across 11 folds. For

Extended Yale B, we will use both random train/test splits and an illumination-based split, where easier lighting conditions are used for training and harder conditions for testing. A separate validation set will be taken from the training data to select model settings such as kernel parameters, neighborhood size, SVM settings, autoencoder settings, dimensionality, and the curvature-gauge threshold. The test set will not be used for parameter selection, and all methods will use the same training, validation, and test splits for a fair comparison.

6. Tools and Techniques

Tools	What we will use it for
Python	As the main programming language for the project
Jupyter Notebook	To run each code step for easy understanding and replication
NumPy and SciPy	For numerical computational work
Scikit-learn	For PCA, Kernel PCA, LLE, Isomap, *k*-means, and spectral clustering
PyTorch	For the autoencoder as a more recent nonlinear method
Matplotlib and seaborn	For plotting figures

How we will judge the groups

Score	Uses labels	Role
Silhouette	No	As an internal cluster quality measure
Davies–Bouldin	No	As an internal cluster quality measure
Calinski–Harabasz	No	As an internal cluster quality measure
NMI (Normalized Mutual Information)	Yes	As an outside check on the groups we find
ARI (Adjusted Rand Index)	Yes	As an outside check on the groups we find

Additionally, the kernel width, σ^2, and the number of clusters, k, will be chosen from the data alone. We will set σ^2 from the median pairwise distance in the space used for that run, either the raw pixels or the PCA-reduced features. We will choose k from the Laplacian eigengap and from how stable the grouping is when the data are resampled. Figures will be made with matplotlib and seaborn, and the numerical results will be saved as tables. The final submission will include the notebooks, a short README, and the package versions needed to run the experiments again.

7. Expected Challenges

The main difficulty is that ORL has far more pixel features than images. That is why a kernel on the raw pictures may not be informative, and it is also why we run the kernel methods twice: once on the original pixels, as the paper did, and once after PCA has reduced the features. Even the second run can depend on how many components we keep, so we will need to check more than one choice of that number.

A second difficulty is that σ^2 and k must be chosen without labels. The original paper never reports σ^2, and the lecture is clear that this setting can change the answer. The median distance, the eigengap, and resampling

stability may not all point to the same value. When they disagree, we will report that disagreement rather than pick the setting that makes one method look best.

Nonlinear methods also take more time and have more settings than PCA. If we tune them more carefully than we tune PCA, the comparison is no longer fair. We will therefore keep the search for settings limited and report running time along with cluster quality.

The curvature gauge adds its own difficulties. Isomap depends on the neighborhood size K and can fail if the neighborhood graph is disconnected, so we will check how stable the gauge is across several values of K and remove disconnected points. With only 400 images in ORL and 165 in Yale, the gap may change between splits, so we will repeat the computation and report its spread. The thresholds for the gap and for "both residual variances high" are chosen on validation data, which may be noisy, and the gauge only compares PCA with Isomap, so it may not predict the behavior of every other nonlinear method. We will report any datasets where the gauge and the observed best method disagree.

Therefore, the internal scores can prefer a grouping that does not match the true number of people. That is useful information for an unsupervised study, so we will report it. Because the datasets are small, a gap that appears on one split may not hold on the next. We will repeat the experiments and use a paired test before we treat a difference as real. Extended Yale B is more realistic, but it is also larger. If time or computing is short, we will finish ORL and Yale first and treat Extended Yale B as extra work rather than as a requirement. Finally, the Yale Face Database is not available from the original authors' repository, so we obtained it from a public mirror on Kaggle. Because this is a secondary source, we will first check that it matches the original release: 165 grayscale images (15 subjects × 11 conditions) at 320 × 243 pixels. We will also confirm that each subject has exactly one image for each condition (center-light, left-light, right-light, with glasses, without glasses, happy, sad, sleepy, surprised, wink and normal), and record and correct any missing, duplicated or mislabeled files. A further problem is that the images have no standard image extensions. Each file is named as the subject ID followed by the facial expression or lighting condition (for example, `subject01.happy`), so the label sits where the file extension would normally be. Inspecting the file headers shows that the images are actually GIF-encoded, and many image-processing libraries cannot load such files reliably. To solve this, we will write a preprocessing script that reads each file's header to confirm its true format and splits the file name into a subject ID and a condition label. The script will convert each image to a lossless format (PNG) under a consistent name such as `subject01_happy.png`. It will also produce a metadata file (CSV) linking each image to its subject and condition, which will later be used to build the training and testing splits. Using a lossless format means the conversion adds no compression artifacts that could affect the dimensionality-reduction results.

8. Deliverables and Timelines

By the end of the project, the following will be completed:

Deliverable	What is included	Due Date
Project Intro Presentation	Introduction to the project and the purpose	October 8th, 13th, and 15th
Demo Code	Initial implementation of unsupervised models	N/A
Midterm Presentation	Progress and recent findings	November 3rd, 5th, and 10th
Final Presentation	Present final findings, discussion, and conclusions	November 24th, December 1st, and 3rd
Final Report	Write about process, findings, discussion, and conclusion	December 4th
Final Code and Notes	Complete code with necessary notes	December 4th

The chart above shows the estimated timeline for the project. The project will begin with initial planning and move to implementation and evaluation, which will lead to final analysis. Although the timeline shows the deadlines as the final dates of each task, the goal will be to complete as soon as possible to ensure time to review, make improvements, and troubleshoot before the final deadlines.

References

[1] W. Huang and H. Yin, "Linear and nonlinear dimensionality reduction for face recognition," *IEEE International Conference on Image Processing (ICIP)*, Cairo, Egypt, pp. 3301–3304, 2009.

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